EXP 5: Comparative Analysis of Different Types of Prompting Patterns

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AIM:

To test and compare how different pattern models respond to various prompts (broad or unstructured) versus basic prompts (clearer and more refined) across multiple scenarios. Analyze the quality, accuracy, and depth of the generated responses.

AI TOOL USED: CHATGPT

PROMPT TYPES:

1.NAIVE PROMPT

2.BASIC PROMPT

1.NAIVE PROMPT:

A prompt that asks a question or gives a task in a **very basic way without specifying details or constraints.**

EXAMPLE: Write about climate change.

2.BASIC PROMPT:

A prompt that clearly asks for a task but with **limited details or constraints**.

EXAMPLE: Explain climate change and its effects on the environment.

TEST SCENARIOS

SCENARIO 1 – CREATIVE STORY GENERATION

NAIVE PROMPT:

Write a story about Mars.

OUTPUT:

A Story About Mars 🚀 🔴

In the year 2075, a young astronaut named Maya traveled to Mars with a team of explorers. When their spaceship landed, they looked out at the vast red landscape filled with dust, rocks, and tall mountains. Everything looked silent and mysterious.

One day, while exploring the surface, Maya noticed a strange glowing rock near a cliff. When she picked it up, the rock started to shine brighter. Suddenly, a small hidden cave opened nearby. Inside the cave, the team discovered signs of ancient water and tiny living organisms.

The scientists were amazed. This discovery meant that life might have once existed on Mars. Maya quickly sent the information back to Earth. The whole world celebrated the discovery.

From that day, Mars was no longer just the Red Planet—it became a symbol of hope and new possibilities for humanity. 🌌 👩‍🚀

BASIC PROMPT:

Write a short story about Mars and a scientist who discovers something unusual on the planet.

OUTPUT:

The Discovery on Mars 🚀 🔴

Dr. Anika Rao was a scientist who had always dreamed of exploring Mars. After years of training, she finally landed on the Red Planet as part of a research mission. The surface of Mars was cold, dusty, and quiet, with endless red sand stretching to the horizon.

One day, while collecting soil samples, Dr. Anika noticed something unusual. Her scanner showed a strange signal coming from beneath a large rock. Curious, she carefully moved the rock and found a small glowing crystal buried in the ground.

The crystal was warm and seemed to produce a faint light. When Dr. Anika analyzed it, she realized it was not like any mineral found on Earth. Even more surprising, the crystal contained tiny moving particles that looked like living cells.

Excited by the discovery, she sent the data back to Earth. Scientists around the world were amazed. It was the first evidence that life—or something similar—might exist on Mars.

Dr. Anika looked across the red landscape and smiled. Mars was no longer just a silent planet. It had a secret waiting to be discovered. 🌌 👩‍🔬

SCENARIO 2 – ANSWERING A FACTUAL QUESTION

NAÏVE PROMPT:

What is fermentation?

OUTPUT:

Fermentation is a **chemical process in which microorganisms like yeast or bacteria break down sugars into simpler substances such as alcohol, gases, or acids without using oxygen.** 🧪

BASIC PROMPT:

Explain the process of fermentation and give two examples of where it is used in everyday life.

OUTPUT:

Fermentation is a biological process in which **microorganisms such as yeast or bacteria break down sugars into simpler substances without oxygen**. During this process, energy is released and products like **alcohol, carbon dioxide, or lactic acid** are formed. 🧪

In fermentation, microorganisms consume sugar as food. As they break down the sugar, they produce different substances depending on the type of microorganism involved. For example, **yeast produces alcohol and carbon dioxide**, while **bacteria produce lactic acid**.

Examples in Everyday Life

1. **Bread Making** 🍞 – Yeast ferments sugar in the dough and produces carbon dioxide gas, which makes the bread rise and become soft.
2. **Curd/Yogurt Production** 🥛 – Bacteria ferment the lactose (milk sugar) in milk to produce lactic acid, turning milk into curd or yogurt.

Thus, fermentation is an important process used in **food production and preservation**.

SCANARIO 3 – SUMMARIZING A CONCEPT

NAÏVE PROMPT:

Explain machine learn.

OUTPUT:

Machine Learning (ML) is a branch of **Artificial Intelligence (AI)** that allows computers to **learn from data and improve their performance without being explicitly programmed.** 🤖📊

BASIC PROMPT:

Provide a short summary of machine learning including its definition, common applications, and benefits.

OUTPUT:

Machine Learning (ML) is a branch of **Artificial Intelligence (AI)** that enables computers to **learn from data and improve their performance without being explicitly programmed**. It works by analyzing large amounts of data to identify patterns and make predictions or decisions.

Common applications of machine learning include **spam email detection, recommendation systems (like Netflix or YouTube), voice recognition, and medical diagnosis**.

The **benefits** of machine learning include **automation of tasks, improved accuracy in predictions, faster data analysis, and the ability to handle large and complex datasets efficiently**. 🤖📊

SCENARIO 4 – ADVICE / RECOMMENDATION

NAÏVE PROMPT:

How to learn skills easily?

OUTPUT:

To learn skills easily, **start with the basics, practice regularly, break the skill into small steps, and apply what you learn in real situations**. Consistency and practice help improve skills faster. 📚✨

BASIC PROMPT:

Explain how to learn skills easily and mention a few useful methods.

OUTPUT:

Learning skills easily requires **understanding the basics, practicing regularly, and using effective learning methods**. Breaking the skill into small steps helps make learning easier. Some useful methods include **watching tutorials, practicing daily, learning by doing projects, and revising what you learned**. Consistency and patience are important to develop any skill successfully. 📚✨